

ActInf Textbook Group ~ Cohort 4 ~ Meeting 14 (Chapter 6 part 2)

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Hey, everyone. Welcome back. It is September 26th, 2023, and we're in cohort four, meeting 14 on our second discussion of chapter six. So before we jump over to the text, does anyone want to just give any thought or take on six? Yeah, briefly, just Charles, it alternates time. So the earlier time, now we're in cohort four, chapter six, and vice versa, but it's all different angles on the same thing. So it's not like there's something that won't make sense in six because... That's cool. I'll just hang around and carry on. Thanks. Yeah. So you want to give a thought on six, or we can look at the questions that haven't been addressed yet. Okay. Let's just go to a question. And then if somebody has another question, they can write in the chat or raise their hand.

Okay. The authors talk about language. But in language processing, one usually assumes recursive processes, e.g. a context-free grammar. How can FEP active inference account for recursivity? In my understanding, hidden Markov models implement types of regular grammars. I assume that hierarchical extensions, i.e. section 642 does not fundamentally change this, does it? Did anyone here ask the question or does anyone want to give a thought on this?

I'm not exactly sure what the question is getting at. However, some senses of recursion can be accommodated by hierarchical modeling. So active inference models of reading have, for example, letters nested within words, words nested within sentences, and so on. So this hierarchical model, embodies the sparsity associated with this kind of a recursive process. And then another kind of related notion of recursion is closer to re-entry, where something is more like self-reflexive. And so that is less referring to the hierarchically nested, like kind of nested for loops notion, and more having to do with re-entry of input and reference, which we might point to, first off, just the fundamental re-entry of the consequences of action through perception via the niche for the agent. And then a little bit more on the sophisticated cognitive entity, the recursive identity concept and like identity re-entering in reference to itself.

Ali, go for it. Yeah.

Ali Kowalikov, Well, perhaps the question also, I'm not sure, but maybe one angle to answer this question is to answer whether generative models are exclusively recursive models or not. So I believe in active inference framework, and especially its current formulation, there isn't any restrictions for generative models to be exclusively recursive. So they can be applied for both, I mean, nested and recursive models. And also in some situations, we can have purely, I mean, non-recursive or sequential models as well. So yeah, it's in its current standing, it is much more broader than only a recursive kind of structure as HHMs that we're dealing with a couple of decades ago.

Great point. And it kind of returns to the theme of chapter six and the questions that are asked in chapter six. It's like, is, or what kind of recursive phenomena is being desired to be modeled?

And then there's sort of like what you intentionally include in your model. Like, you know that you want to have this phenomena and you're going to engineer and implement motifs or mechanics to get this phenomena. Like it's your target phenomena. You want to have a phase transition associated with people like leaving a crowded room. And then like, you want to like, kind of like build that into the model super explicitly. And then a little bit more indirectly would be you want to have some phenomena arising, but you don't want that phenomena to be like explicitly a parameter in the model.

Um, but that's the chapter six question. What do you actually want to model? What system of interest do you want to model? And, and that's non-trivial to identify the boundaries of the system and to subset what you actually want to model about it. And what's interesting, what questions you want to model? Because otherwise you're just making a map that's the territory and there's no point in doing so. Okay. How's the active inference modeling recipe or approach to modeling similar and different from other modeling approaches? That's a very good question.

Just from what people have seen or what they feel, what's, what's one or more similarities or differences, whether people have, have done some quantitative modeling like statistics or systems engineering, or whether they've done some other informal analysis approach, how is, or could be active inference model? And what's chapter six recipe different? Yeah, Ali.

Well, I think this question, um, is, um, I don't know, perhaps too general, uh, to be answered, um, sufficiently or helpfully, but, um, uh, when it comes to active inference modeling, uh, my own understanding is that, uh, the main thing that sets it apart from, uh, perhaps all the other, or many other approaches, uh, uh, is, um, its emphasis on constructing, uh, the useful and appropriate generative model. So, uh, in all the other approaches, uh, the, uh, the representation of the world or the phenomena, uh, we're trying to model is not something, uh, that, uh, it's not, uh, uh, on, uh, very clear, uh, probabilistic representation terms, uh, uh, so to speak, but in active inference, uh, there is this clear distinction between, uh, what, uh, what kinds of phenomena or, uh, what, uh, maybe we can say what kind of structure, uh, the phenomena or, uh, uh, uh, uh, the, uh, the system we're trying to model encapsulates in terms of its, uh, probabilistic behavior, uh, so, uh, it is much more clear than other modeling approaches, uh, uh, when it comes to the representation, uh, of that a specific system world or, uh, the phenomena of interest, uh, uh, uh, in relation to, uh, its generative model construction. Uh, so, yeah, I think, uh, the main distinction, uh, lies exactly in, uh, the approach to constructing that representation.

Thanks.

Anyone else?

Feel free to raise your hand or go for it.

I'll, I'll also point to recent work by now at all from this year, where they, uh, designing explainable artificial intelligence with active inference, a framework for transparent introspection

and decision-making.

It points towards some of the motifs that make it different and which give it a semantic interpretability because when features of the system, including its own introspection, its own confidence on its answers or its sense-making are framed in terms of probabilistic parameters, that are just what it says on the label.

Then there is a clear sense of what structure or phenomena arising from structure of the system that we're modeling.

That doesn't mean that the model in hand is going to be the system we're modeling.

Obviously that's the whole map territory fallacy, map territory fallacy fallacy.

So, um, it's not a clear, um, situation, but when we have, um, interpretable parameters that play explicit roles in semantic Bayesian networks, then, um, at least we're on that path.

Yeah.

I'll, I'll, um, I'll bring that paper into, I don't, I'm sure it's somewhere else in here, but, um, Charles, this is the paper of, uh, Ramstead et al.

On the map territory fallacy fallacy.

And just in brief, commonly people talk about map territory fallacy, which would be the fallacious equivalence of a map in a territory.

People are pretty familiar with map territory fallacy map.

It's not the territory, all of that, you know, maps are useful.

Some are all of that.

And in this paper, they point to a kind of extreme case where people will look at active inference, free energy principle modeling and say, oh, well, it's just making a map.

So it's, it's just making a map of the territory.

So it's not the territory.

So it's kind of like, what's, what's the issue?

Like what, what's the value add?

Um, and so, uh, there's a few aspects to this and, uh, it relates to on one side, um, reification of the map and the territory.

Like you go with the map to the territory and then like, you, you, you know, that it's, that the map's not the territory yet still they are used to kind of co justify each other.

And then another flavor is, oh, well then free energy principle does nothing.

If it's just helping us build maps, that's nothing.

That's just a mapping framework.

And that's like a fallacy on the map territory fallacy.

Oli or, or anyone else?

What, what, what other angles are there?

I can actually just chime in quickly.

Cause it relates to the one or sort of two part question comment that I did put into Coda last week, um,

relating to chapter one, but it had, it may be, it's another angle on the same thing, which is the modeling versus action.

It's kind of a, it's kind of a, a can of worms, but I don't know if you, if that's enough of a prompt to kind of go for it.

Yeah.

I don't know if it's terms.

Yeah.

How do you see it?

Well, I, I'm not sure.

This is just, was my, one of my initial questions that came up right away.

Um, where, I guess the answer is in the definitions, but, but in practice, how is the modeling, modeling as action compared to modeling in contrast to action?

I guess that's the best I can put it right now.

Okay.

That's very interesting.

Like modeling is an action.

It's an iterated process that a modeler takes.

In fact, we could even do a chapter six on chapter six and make a cognitive model of somebody applying active inference to modeling.

Eventually I expect that that would be relevant.

So there is a sense in which modeling is an action.

It's a practice.

It has explicit and implicit skills associated with it.

So modeling is action.

However, modeling, throwing a baseball isn't throwing a baseball.

There's a tiny bit more I can add, which I see I had written also, I think in Coda, um, the cohort five of, of course, observing and documenting are both also action.

However, the systems and behaviors being observed would be the primary actors in the sense of action.

Yeah.

Yeah.

Anyway, it has to do with that.

Observing as action as it relates to that, too.

Thank you.

Yeah, we'll, we'll, um, return to these, but, but, um, think about the setting of, of, so let's just say we're chapter six in a baseball game.

So we have a person who's a spectator.

They are still taking action with their eye.

So, okay.

To look at different parts of the field, reducing their uncertainty.

And then covertly internally, they are deploying attention as action, attending to different parts of their experience.

So even a seemingly passive, perhaps overtly passive observer.

Is active at least in the attentional covert internal sense, if not also in the epistemic foraging.

I secade sense like choosing where to look, but none of that activity happening in the spectator is the action happening on the field.

And then to connect this back to the map territory fallacy fallacy, no matter how deep we go into this, um, model of like a spectator and an athlete.

It does the, the end of the process isn't going to be like a material spectator and an athlete.

So, um, the map of that setting, that cognitive setting isn't the territory.

That's so, so now, now, now we're immune to the map territory fallacy.

Map is not the territory, but then what is it?

It's a map.

And then, um, I'll also note that, that Ramstead and Saktiva de Vel and Fristin go a little bit further.

And then, um, just naming, um, their critique of this deployment.

They come back even stronger and say, actually free energy principle is the ideal model.

And any model that doesn't conform to the free energy principle with respect to the map territory scenario is either saying too much or too little.

Ali.

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Ali.

When it comes to map territory fallacy fallacy, my understanding is that the FEP and active inference escape this exact meteorological fallacy right from the beginning by not conflating map and the territory.

Or in other words, by taking a totally different approach to the concept of agency.

And it's not something that's just metaphorically ascribed to part of the system.

So when they say that a part of a system looks as if it does some kind of inference or any action, it's not just a pure metaphor.

It's built into the very fabric of active inference theory.

So, yeah, I believe in that case, this kind of self-referential scenario can be applied indefinitely if you wanted to with this specific kind of approach.

Thank you, Ali.

I'll see if this is kind of another piece of it.

When we say a system looks as if it is bounding surprise or it's reducing surprise, that's not metaphorical or the realist claim about the territory.

It's a narrow and limited claim about the map.

And now all claims about maps, unless they're being considered in some abstract mapping setting, are on the foundation of the relationship between the map and the territory.

How is it being approached differently in ActInf?

We have figure 9.2.

9.1.

The meta-Bayesian.

So here in the center, we have the cognitive model of the system of interest of the subject of the modeling subject.

And then outside of the subject is this figure and ground.

So we're being explicit that we're making a map about the cognitive model.

Of the subject.

Now, that subject's cognitive model could be a map of the environment as well.

But once we enter into this mode of modeling, we're not making claims about the subject.

We're either making claims about our map of the subject.

And then a subset of that is maps that the subject makes about the environment.

I hope that didn't say too much or too little.

But I think that is what resonates with me when Alisa says that we kind of head off that fallacy from the beginning.

Because the task that we're doing in cognitive modeling.

If this is a rat in a T-maze or if this is some digital environment or something like that.

Is we set up our work explicitly as making a map.

Rather than like, well, what does the amygdala do?

It's like, well, then you're going to get into a situation.

It's like, are we making claims about the amygdala?

Because that's a claim about the territory.

But if we say, well, we're building a map of the amygdala.

And one of the functionalities in our map is that the amygdala is making a map of its environment.

So that's the kind of nested mapping.

And that may relate to the earlier recursion question.

But now if we have prevented or stymied off this map territory fallacy, which is actually fairly easy to identify if it's approached hygienically.

Then we're just not making claims about the amygdala directly.

We're building the map, which can include submaps.

But then there doesn't have to be.

I mean, this is the Scylla and the Charybdis.

Map territory fallacy on one hand.

And then the fallacy fallacy on the other hands.

Because like the map's utility and relevance is related to its mapping.

Of course.

Which we want to be able to say things like, yeah, 4th Street and 5th Street run in parallel.

And I'm not just talking about my map.

But that's the empirical question.

Okay.

I had a question about the map and territory.

Could you explain more about what the fallacy is?

Yeah.

I'll give a first take.

Broadly, the map territory fallacy is when a constructed map, which is to say an abstraction or a model of some real material system of interest or territory, when especially in a science or research or modeling setting, it can be easy to kind of conflate or use rhetoric from one regarding the other.

So we have a bunch of healthcare data on doing this and this health outcome.

And we did a linear regression.

And we found that smoking was increasing the risk of lung cancer.

Sounds like a relevant thing to say.

But the more narrow statistical claim would be, there was a statistically significant positive regression slope between this measure and that measure.

So that's on the mapping.

We did a linear regression.

And we want to be able to, I mean, it's like, look at the line.

Don't we want to be able to say that there's like a positive relationship between these things in the real world?

So that's the tension.

Because you'd want to be able to just, first off,

look at your statistical results on the map

and make a statement about the territory.

Like doing this increases that.

But that's already leaving the map.

It's like, you're looking at the map.

It's like, well, it looks like these two roads intersect at this GPS coordinate.

But it's like, but that doesn't mean that they do.

Especially if you're like extrapolating.

So then if somebody were to have smoked twice as much, then they would have this much more.

So the map territory fallacy is just,

it's done in science communication.

It's done in self-talk.

It's done in so many different settings

where something is gleaned or constructed in the map, as all maps are.

And then that is used to justify

and sometimes even linguistically is woven

or completely mixed into claims about the territory.

Which may be valid or aligned or resonant

or like pointing in the same direction.

So it's not like we're not in a situation

where like maps are like deceptive.

Unless they are, but that is the tension.

What can you say about the map and the territory?

But it starts with just the fundamental recognition

and the repeated exhortation

that the map's not the territory,

which is why you hear that,

as well as like all models are wrong,

some are useful.

Or the Borges story of the map maker
who like makes a map that's so detailed
that it covers everything.

And then like it becomes the territory.

Those are all kind of allegorical ways to talk about it.

But the big issue is

when we're doing scientific modeling,
we're doing map making.

And there's this complexity of relationship
between any map and any territory.

And again, it would be so simple
and straightforward to just make the map,
do the statistics,

and then the statistics should just apply
right back to the territory.

You know, I have GPS locations on these two objects.

I made a statistical model of their location.

In the statistical model, they're close.

So can we say that the objects are close?

It's like, just let me say that.

But if it were that way,

yeah, exactly.

As Charles says, it's not so clean and distinct.

There are settings where it is clean and distinct,
but those are not every setting.

That makes sense.

And it happens a ton in neuroscience as well
because the territory is so messy and ineffable
that it's like somebody will make some list of terms
or they'll make some axes that they're studying,
summary statistics or personality traits
or just somebody brings some framework to reckon
or to grasp with neuroscience or with psychology.

And then they start reifying the existence.

It's like, I can't believe there's things
in the upper left quadrant of this thing that I made.

But that's, again, it's a statement of maps,

not territories.

Ajith?

Or Ali, sorry, go ahead.

Yeah, I just wanted to mention

a probably useful distinction made by Deleuze

between representation and cartography.

So he makes a clear distinction

between what we want to represent,

in other words, what we want to trace in our models,

and the cartography or the action of mapping something in the model

or diagramming or metamodeling or whatever.

So in this sense, I think what active inference does,

I mean, in light of this distinction between representation and cartography,

more maps into the representation side of modeling.

In other words, the very active constructing a generative model of the world or the phenomenon of interest

is done by tracing the probabilistic states of the world.

So it's not cartography in the sense of mapping or diagramming or metamodeling,

but rather it actually does this very active tracing representation.

So I think it also helps to distinguish between the usual way of talking about mapping or modeling in other scientific endeavors

and the active inference way of doing things.

Would there ever be a case where you are talking about the territory?

Like, I'm just kind of wondering, like, when you would ever talk about the territory in science?

Because wouldn't it always be sort of some representation you're working with?

What do you think?

Oli?

Yeah, so it depends on what we mean by

representation in relation to the territory we're trying to, quote unquote, represent.

But, I mean, even the simplest act of observing the world and observing anything that we want to model

entails

at least

at least

at least

an elementary kind of

kind of

representational cartography

kind of representational cartography

kind of

kind of representational cartography combined.

So,

maybe, I mean, it's not possible theoretically and philosophically even to

have a direct access to the territory we want to study.

But the only way we can have anything useful to say about that specific territory is through the representation and cartography.

At least this is one of the views held by structural realists, and specifically ontological structural realists.

So, it's not about the content of the world, but the structure, the structure of the content of that world manifest themselves that scientists are interested in.

So, yeah, at least in in some perspectives it's not possible to have that direct knowledge of the territory.

I think it's an excellent question as many we just kind of surface them, and it's for people to consider. But, but.

If we were engineers, and if we were building a bridge, and we had the blueprint, and we were talking about the blueprint, like, is it sufficient to just make strong claims about blueprints that we have in hand on the table, and then

understand that there's a relationship between the blueprints that we make, and the territory that we care about.

Is that enough? Like, is that like a firewalled approach?

We're, we're, we're only going to talk about the blueprint that we have on the table, and blueprints are related to bridges.

Or are there times where we want to bypass the blueprint, and just say, you know, maybe look at the bridge, and say these two things look like they're this way.

Um, and then yeah, what do you think about this?

Yeah, like, I guess there are some cases where you would have to make sure you know that they're distinct.

Um.

Especially like when you're considering the things the blueprint can't really capture about the bridge.

Yeah.

And in, in neuro and psychology, these, these come up, especially like when classifying brain regions, or when classifying, um, different neurodiversities.

And there's one topic, um, that, that, that.

All these mentioned of the traces.

In the world, this reminded me of Bruno Latour's work on actor network theory and ants.

And this is a very like proto act imp approach.

It's all about resuming the task of tracing.

And it's, it's very provocative and another, um, I think provocative philosophical, um, term in this, uh, space is always already.

I think like, these are just two English words.

Come on now.

The perception of phenomena by the mind of an observer, the features of phenomena that seem to precede any perception of it are said to be always already.

Present.

That is very provocative.

Provocative.

I believe.

Or evocative.

It's like.

It was already that way.

So.

We're showing up in the stream of things.

Potentially resuming the tracing.

Of associativity.

In a map making setting.

There.

I was being completely exoteric.

I really was talking about.

A.

I was being completely exoteric.

I really was talking about.

A.

So.

I was being completely exoteric.

A.

[illegible]

about a bridge that like cars would drive over.

So compared to the actual bridge then?

Yeah.

And then a neuroscience setting,
you know, we'd have,
we have the person's body is the territory.
Okay, cool.

And then, yeah.

And then the blueprint would be some,
some, you know, chapter five,
like, well, here's how, you know,
the elbow bones connected to the knee bone,
but the body is a complex territory,
which is why so much has been written and explored on it.

Indeed.

Thank you.

Yeah.

Cool.

And the body's a bridge too, right?

Sorry.

Go ahead.

And the body's a bridge and the body's body as map body as void,
you know, this, as that, this already always as that,
like that's, that's the fun.

Okay.

Let's go to this question from Andrew.

Okay.

Self-talk is one of the topics or activities where map territory concerns apply.

Any direct work or paper references on this topic you could share?

Hmm.

What do people think?

Just to chime in.

I mean, I, I just heard you say that along the way,
whenever we were discussing the fallacy there and it just really piqued my interest.
So, but I mean, I can already imagine how that could play out as far as someone considering,
you know, the self, the cognitive process of like, they're attempting to map whatever it is,
that's going on with them, whatever it is they're directing their attention towards versus the territory.
That is an entire generative process going on.
And probably from an active inference framework,

they're only going to be able to kind of approximate in their own model of that territory.

But in any case, I just, it's so, I feel like I could get into an infinite regress here,

which is thinking of what's going on whenever I self-talk and, uh,

just curious if there was any published work here or, um, anything along those lines.

I don't know of any published work on this.

I'm sure there are analyses on like self-reported self-talk,

but it's something that is understandably hard to measure in flow.

And you get into all these kinds of like recursive questions.

What if you remembered saying it to yourself, but you didn't in the moment,

or you did say it in the moment, but you didn't remember it or tend to it.

Um, yeah, I, I mentioned it just because it is like

the kind of singularity of map and territory.

Like, okay, let's just say that the, that a cognitive activity is making maps.

So, um, just, you know, putting ourselves in the driver's seat, not figure 9.1,

not behavioral researcher, but you could take that stance too, but view from the inside.

We are making maps of our territory, you know, where our objects are in the room and all these other things.

And then that map re-enters

through heard language and sub vocalized language.

So it is embodied action as well.

Ajith?

So by self-talk, do you mean like an inner monologue?

Um, I think I might be missing some context.

Yeah.

Yeah.

Like stream of consciousness, whether they just.

The, the, the internal flow and imagination.

Um, I've seen a paper published where they use.

Like fMRI data to predict what people's inner monologues are.

If that might be something.

That could be some way of mapping out this territory.

Um, also what people are sort of imagining visually and also what they're listening, like what they're perceiving in terms of music.

Things like that.

Um, I think like the latest sort of deep learning models are actually getting sophisticated enough where they can be able to predict what people are intuiting.

Just using like the right type of data.

Um, just to say that very much is in line with the kind of thing I'd be interested in, uh, I guess, bonus points that I'm interested in neuroimaging data, especially.

So, um, I just shared is, is this, um,

I can send some papers that, uh, uh, actually.

Yeah.

Put them in here.

Just follow the little token and then add them in here.

Okay.

Oh, I, I just shared one just to see if I could confirm with you is, is this at all, um, in line with what you were referring to there?

Uh, is it on the screen?

Oh yeah.

Yeah.

That is, uh, pilot fMRI data.

Um, um, quick Google search.

I'm not trying to throw us off track here.

Uh, this is not the one I was referring to, but I can find the one I was referring to.

I think it was like recent.

It was like in 2023.

Oh, that's even better.

Right.

Great.

I really appreciate it.

Thank you.

Wow.

Yeah.

It's, this is very, I mean, and to kind of return to chapter six, like this is why we have a chapter six.

This is why unified approaches to perception, cognition, action matter.

This is why having an active inference ontology is important.

This is why having a recipe for making generative models and having some of this meta theory or meta position on the relationship of map and territory fallacy and fallacy fallacy and fallacy cubed.

Um, because these are not just massively interactive and nonlinear systems of interest.

They're also ones that were always already experiencing.

So if it was just a Rube Goldberg machine, that would be one thing, but it still wouldn't even be of the type.

Qualitatively.

Qualitatively.

Of.

Without even going too deep into phenomenology and like experience and awareness.

It just wouldn't necessarily be of the type.

That would be a recursive re entrant.

Cognitive system.

And this is why, um, I think the textbook ends.

With.

These lines.

Important rights of passage in theoretical neurobiology.

Are trying to write down a generative model.

Experiencing the frustration when simulations misbehave and learning from violations of your prior belief when something unexpected happens.

So it's like.

This isn't just a hot take.

Like there's.

There's.

What can you say about some of these settings?

That's just like, oh, okay.

It's.

It's.

You don't just like.

Drop the mic.

And like dust off your hands.

In this.

Iterated.

Generative modeling.

Practice.

It's.

It's like.

Especially if we're.

Applying it to a setting like self-talk, let's say, or some other psychological phenomena.

Which, you know, impinges or overlaps on self-talk.

You write down the generative model.

There's a recipe.

It's chapter six.

Plus some of the other pieces that people have added in.

You write it down.

You roll it out.

You see what happens.

Did you expect what happens?

Well, if you understand act inf and you made a simple example.

You may have.

You may expect.

Get exactly what you expect back.

Like.

Well.

I made it.

I made an oscillating pendulum.

And it did exactly that.

It's like.

Great.

Well, that might be like a homework assignment for like.

A certain.

Course.

But.

If we're interested in something with more richness than like a pendulum.

Then.

When we unroll.

That.

Generative model.

We will be surprised.

And then.

What do we do with surprise?

Well, we know there's two ways to bound it.

Change mind.

Change the world.

Change your mind.

Learn an update.

Change the world.

Re-enter.

Change the generative model itself.

On the table.

So.

Having a structured.

Approach.

Profession.

Ontology.

Methodology.

Toolkit.

Etc.

Etc.

Comes into play.

And chapter six.

Is kind of the.

Transition point.

Or like the fulcrum.

That the book is actually organized around.

Chapters one through three.

Get us.

To active inference.

Context.

Low road.

High road.

To.

Or of.

Chapter four.

Generative models.

Themselves.

Chapter five.

Generative models.

In the mammalian neuroscience.

Setting.

And then.

Pretty much as soon as we can.

Chapter six.

Now it's us.

Making that generative model.

And.

Even.

Without.

Extensive.

Math.

Or programming.

Experience.

We can start to address.

Some of these questions.

And then we're already.

In the flow of things.

It can definitely feel like.

Chapter four.

Is like.

Well.

Chapters two and three.

Can feel like.

Waiting.

Through like.

Every single department.

Of.

Learning.

Just to get to.

Through.

And then chapter four.

Can feel like.

Getting through.

Is like.

Impossible.

Slash.

Improbable.

All.

Totally.

Make sense.

In terms of their evaluation.

And how they're experienced.

But then like.

We break out.

On the other side.

Of four.

Just from a page turner.

Perspective.

In six.

And this is like.

Such an incredible.

Open.

Conversation.

About.

Modeling.

And these.

They're hard questions.

To.

Talk about.

But.

Hopefully.

People can see.

That at the.

Single entity.

Or at the.

Research group scale.

Having.

Alignment.

Or shared.

Reference frames.

Around some of these topics.

Are make or break.

Thanks.

If you can add.

And just.

Just.

You know.

Add.

Make little sub pages.

Or add your own question.

Or just.

You know.

Just.

Just.

The more references.

And the more things.

People add.

The better.

It's awesome.

Okay.

Yeah.

I can do that.

Thank you.

One other piece.

I'll just add on this.

Kind of map territory.

Is.

The.

The digital.

Introduces.

A very interesting setting.

That's kind of like a chess board.

In the sense that it's fully observable.

And that these are virtualized.

And that these are virtualized.

Settings.

Where.

Strong concordances between maps and territories.

Can be constructed to exist.

So.

Like.

Like.

There can be a schema.

An abstract.

Pattern or scheme.

That.

Then.

Generates the territory.

In a.

We know.

Ultimately.

It's embodied.

Through like.

Thermodynamics.

And information.

All of this.

So.

So not.

But.

But to a first approximation.

The digital systems.

Have a different map territory relationship.

Than.

Birds.

To the inner talk point.

Body's eye.

This is just making me think like.

Inner talk.

Is.

Is.

Re-entrance.

Auditory.

Action.

And perception.

Even though it doesn't go through the ear.

I assume that these.

Imaging studies.

Ask the question.

Maybe even find the result.

That there's like activation of.

Of auditory processing.

Or something like that.

Then we have visual.

In a.

But these are.

So those are.

Internalized.

Exteroceptive processes.

Inner eye.

Inner ear.

Inner touch.

But then like.

Do we have.

Internalized.

Interoceptive.

As well.

Like.

Can we take an inner.

Virtual view.

On something.
That's already inner.
And.
Is that.
You know.
Different.
Where is it more or less.
Relevant.
Than.
Self-talk.
But if you could take.
An inner.
Position.
On.
The interpretation.
Of.
The self-talk.
Then I think.
We get into.
Andrew's.
Infinite.
Regresses.
Yeah.
I don't know.
It just brings up.
So many.
Questions for me.
Of how to.
Conceptualize that one.
Because I mean.
In a certain way.
If someone were.
To engage in self-talk.
I could view it.
As like.
They're.
If we view the.
Self.

Itself.
As like.
A kind of map.
That the.
The person.
Has constructed.
So you're.
You're.
You've taken your map.
And now you're.
Making a map.
Of your map.
Perhaps.
Like analyzing it.
Or maybe you're just engaging.
In some kind of.
Nested.
Structure.
Learning.
Like.
Oh.
Let me.
Revise.
My.
Model of.
Myself.
I don't know.
There's so many.
I think.
You know.
Some kind of.
More empirical work.
Would need something.
A little bit more.
A finesse.
Than the way I'm talking.
About it right now.
But it's.

Very interesting ways.

It could go.

I could imagine.

I don't know.

I don't know.

Yeah.

Like.

The system can't.

A system cannot.

Just.

Provide.

If it.

If it just.

Represents.

Everything.

Then.

It's just a pass through.

System.

There's no attention.

Or.

Distillation.

Or representation.

Just a straight.

Pass through.

So.

Whether we're talking about.

Our selves.

Or some other.

Hypothetical system.

The.

Self representation.

That it explicitly.

Makes to itself.

Is.

A sample.

Or a selection.

Or an attention.

Distribute.

Reallocation.

Of.

Something else.

Like.

That's just like.

Kind of.

Light and dark.

Like.

Is it all.

Equally lit.

And it's just a pass through.

Or.

Is there going to be.

Some area.

That is.

Brought.

More into focus.

In.

The.

The.

Representation.

And then.

So.

What.

Should be.

Re.

Presented.

Maybe.

That's where people.

Have different.

Views.

Obviously.

But.

There's.

The preference.

To have.

Our.

Representation.

Be.
Amicable.
Or like.
Pleasant.
That's clearly.
One drive.
Another drive.
Is.
For meaning.
All these.
Different things.
And.
I think.
You recall.
Somewhere.
Earlier.
In the textbook.
Tristan.
Well.
The author.
To say.
Something.
About.
An.
Agent.
Having.
A.
Model.
Of itself.
In order.
To differentiate.
Itself.
From.
Its.
Environment.
I.
E.
The.

Generative.

Process.

Around.

It.

And how.

That may.

Be an.

Essential.

Possibly.

An.

Essential.

Aspect.

At least.

For certain.

Entities.

In.

Maintaining.

Something.

Along.

The.

Lines.

Of.

Homeostasis.

That is.

If you don't.

Have.

A sense.

Of.

What you need.

And what you need.

To do.

Which is part.

Of.

You.

And.

Potentially.

Dissipate.

You know.

To.
The entropic.
Processes.
Around.
You.
So.
Anyway.
I don't want.
Keep.
Yeah.
But.
It's interesting.
Yeah.
It's fun.
Yeah.
As.
As usual.
Another.
Great.
Chapter.
Six.
Journey.
Chapter.
Chapter.
Six.
Is.
Is.
Like.
An inflection.
Point.
In.
The.
Book.
And.
And.
And.
How.
We.

Approach.

Things.

So.

We.

Will.

Then.

In.

The.

Coming.

Weeks.

With.

Alternating.

Times.

Head.

Into.

Chapter.

Seven.

And.

Eight.

On.

Discrete.

And.

Continuous.

Time.

Models.

Chapter.

Nine.

On.

Data.

Based.

Analysis.

Chapter.

Ten.

Wrapping.

Things.

Up.

All.

Right.

Thanks.

Everyone.

Till next time.

Yeah.

Bye.